Cooling Load Chart

A heat load in kilowatts expressed as BTU per hour and tons, and the airflow and chilled water flow needed to carry it away.

One kilowatt of electrical load is one kilowatt of heat to remove; in US units that is 3,412 BTU per hour, and a ton of cooling is 3.517 kW. Nearly all the electricity an IT load draws leaves as heat, so a 100 kW room needs 100 kW of sensible cooling, about 28.4 tons, before any allowance for other loads or redundancy.

Heat load

kW	BTU/h	Tons
1	3,412	0.284
5	17,061	1.42
10	34,121	2.84
20	68,243	5.69
50	170,607	14.2
100	341,214	28.4
250	853,036	71.1
500	1,706,071	142
1,000	3,412,142	284
2,000	6,824,284	569
5,000	17,060,710	1,422

Sensible heat. For IT equipment, the electrical load in kW is the heat load in kW.

Airflow per kW

Air temperature rise	m³/h per kW	L/s per kW	m³/h for 100 kW
5 K	596	165.7	59,643
8 K	373	103.5	37,277
10 K	298	82.8	29,821
12 K	249	69.0	24,851
15 K	199	55.2	19,881
18 K	166	46.0	16,567
20 K	149	41.4	14,911

Standard air at sea level. A wider temperature rise across the equipment needs less air. One kelvin is the same step as one degree Celsius.

Chilled water flow per kW

Water temperature difference	L/s per kW	L/s for 100 kW	m³/h for 100 kW
4 K	0.0598	5.98	21.5
5 K	0.0478	4.78	17.2
6 K	0.0399	3.99	14.4
7 K	0.0342	3.42	12.3
8 K	0.0299	2.99	10.8

Water temperature difference	L/s per kW	L/s for 100 kW	m³/h for 100 kW
10 K	0.0239	2.39	8.6
12 K	0.0199	1.99	7.2

Plain water. A wider difference between supply and return needs less flow.

How the values are calculated

$$m^3/h = kW \times 2{,}982 / \Delta T\ (K)$$

$$L/s \text{ of water} = kW / (4.18 \times \Delta T\ (K))$$

$$\text{Tons} = kW / 3.517$$

Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

These are design-point figures for sensible heat. They leave out lighting, people, heat gain through the building, humidification and the cooling plant's own losses, and they include no redundancy. The air constant of 1.08 applies at sea level; at altitude more air is needed. The water constant applies to plain water, and a glycol mix needs more flow.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/cooling-load-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.